

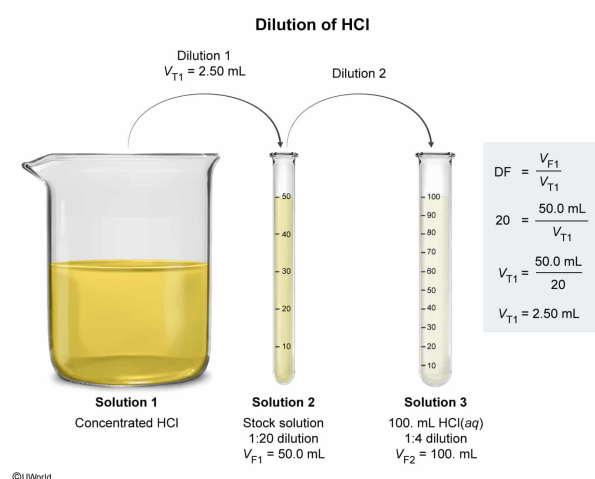
A sample of concentrated HCl (Solution 1) was diluted 1:20 to make a 50.0 mL stock solution (Solution 2). A portion of Solution 2 was then diluted 1:4 to make a 100. mL aqueous HCl solution (Solution 3). What was the volume of the sample of Solution 1 diluted to make Solution 2?

- ☐ A. 0.250 mL [5%]
- ✓ ☒ B. 2.50 mL [70%]
- ☐ C. 25.0 mL [18%]
- ☐ D. 250. mL [6%]

Correct

69% answered correctly

Explanation:



Concentrated solutions may require **dilution** if the amount needed is smaller than can be accurately measured. The concentration of the solution is reduced by dissolving a small volume in a larger volume of solvent (V_S). The **final volume (V_F)** of the diluted solution is the sum of the volume transferred (V_T) and V_S . The **dilution factor (DF)** is the magnitude by which a solution's concentration is decreased and is expressed as the ratio of V_F to V_T . The DF in a multistep dilution can be calculated using the V_T and V_F of each dilution:

$$DF = \frac{V_{F1}}{V_{T1}} \times \frac{V_{F2}}{V_{T2}}$$

Two sequential dilutions were done to make a 100. mL aqueous HCl solution (Solution 3). A sample of concentrated HCl (Solution 1) was diluted by a factor of 20 to make a 50.0 mL stock solution. The volume of the sample of Solution 1 can be calculated by plugging in the DF and the V_F of the solution from the first dilution into the DF equation:

$$20 = \frac{50.0 \text{ mL}}{V_{T1}}$$

$$V_{T1} = \frac{50.0 \text{ mL}}{20} = 2.50 \text{ mL}$$

Therefore, 2.50 mL of Solution 1 was diluted to 50.0 mL by a 1:20 dilution.

(Choice A) If V_{F1} were 5 mL rather than 50.0 mL, a V_{T1} of 0.250 mL would be obtained.

(Choice C) The volume 25.0 mL is the amount of the stock solution (Solution 2) diluted to [make Solution 3](#).

(Choice D) A volume of 250. mL would be obtained if 50.0 mL were divided by a DF of 0.2 rather than 20. This volume is not plausible for V_{T1} because the V_{F1} is 50.0 mL and the volume transferred cannot be larger than the final volume of the solution.

Educational objective:

Dilutions are performed to decrease the concentration of a solution when a small amount is needed and cannot be accurately measured. A small volume of concentrated solution (V_T) is transferred to a certain volume of solvent (V_S), and the final volume (V_F) is the sum of V_T and V_S . The dilution factor can be determined by the ratio of V_F to V_T .

Time spent: 0 sec

Subject:
Organic Chemistry

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System:
Separation Techniques, Spectroscopy, and Analytical Methods